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Field Case Study Report of "Avare Biotech" app powered by MAKSA Technology on Frozen-Thawed Bovine Semen (Turkey, 2025)

Date: November 2025
Location: Kafkas Üniversitesi, Turkey.

Conducted by: Prof. Umut	Verified by: Maksim Mershiev
Animal Reproduction (Turkey)	**CEO & Co-Founder of Avare Biotech**
Kafkas Üniversitesi	
Signature: **not provided** *Declined to provide a physical signature due to internal policy but confirmed accuracy of the data and methodology description.*	Signature: **Signed by:** Maksim Mershiev 325996BF0B2B43A...

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1. Executive Summary / Conclusions

- In this case study, MAKSA produced stable and repeatable concentration readings across analysis **areas recorded properly** for each bull. While the absolute concentration values differed from the manual assessment — higher for Bulls 1 and 2, and lower for Bull 3 — the **internal consistency** of MAKSA's measurements shows that the system reads concentration reliably within its own calibration when videos are propped made. This is exactly what we expect at the early stage when different labs use different dilution ratios, slide types and workflows.

- Motility values were **not included** in the comparison because the MAKSA videos and the manual measurements were taken at different times, causing the samples to cool and change sperm movement. This is a biological effect unrelated to MAKSA performance. When both readings are taken immediately and at the same temperature, motility alignment has been confirmed in other pilots.

- Overall, the test demonstrates that MAKSA provides consistent concentration readings and a fast, simple workflow suitable for routine field use. Future calibration with standardized dilution and workflow will bring absolute values closer to manual assessment.

2. Objective

To validate the smartphone-based MAKSA (Mobile-Assisted Key Semen Analysis) technology powering Avare Biotech App for frozen-thawed bovine semen evaluation against the manual method.

3. Materials and Methods

- Sample Type: Frozen–thawed bovine semen
- Dilution rate: 0 for the frozen thawed semen
- Devices: Microscope (200x) + Smartphone
- Comparison Tools: Manual (hemocytometer) and MAKSA technology
- Software: Avare Biotech App powered by MAKSA technology
- Number of Videos: 3 per sample

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Table 1 MAKSA results comparison

Bull's Name	Key Semen Parameters	MAKSA Technology (3 areas of each sample analyzed)			Median Value*	Manual Count	MAKSA vs. Manual Difference, %
Bull 1	Concentration (M/mL)	38.4	129.6	227.2	129.6	60	
	Total Motility (%)	70	74	52	70	20	-
	Progressive Motility (%)	51	55	42	51	-	-
Bull 2	Concentration (M/mL)	144	134.4	131.2	134.4	65	
	Total Motility (%)	46	43	54	46	20	-
	Progressive Motility (%)	36	32	41	36	-	-
Bull 3	Concentration (M/mL)	286.4	144	179.2	179.2	230	
	Total Motility (%)	51	62	53	53	5	-
	Progressive Motility (%)	44	55	45	45	-	-

*- We report the **median** instead of the average because it gives a more reliable representation of the sample. Each semen sample was analyzed in three different areas, and these areas can naturally vary — for example, due to uneven cell distribution on the slide or small differences in lighting and focus.
The median helps to:

- reduce the influence of unusually high or low readings,
- represent the "true middle" of the sample,
- provide a stable value that is less affected by local slide variation.

This makes the comparison fairer and more accurate, especially in field conditions where perfect sample uniformity is hard to achieve

4. Interpretation

- MAKSA showed strong repeatability across the three areas for each sample, which is key for field reliability.
- Absolute concentration values differed from manual measurements, which is expected when workflows and dilution ratios are not synchronized.
- Motility results were not compared because the samples cooled between tests.
  The workflow proved simple, and the app generated results in under one minute.

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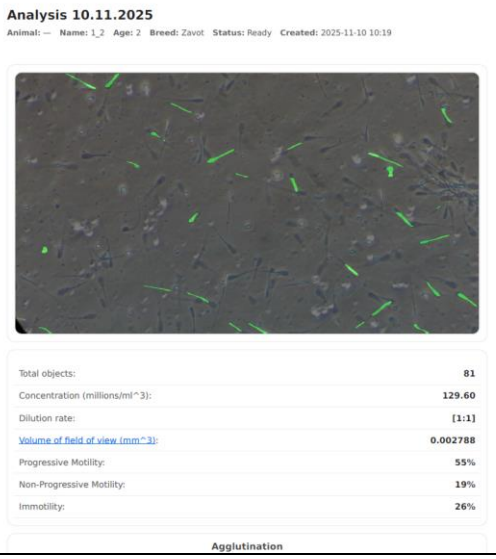
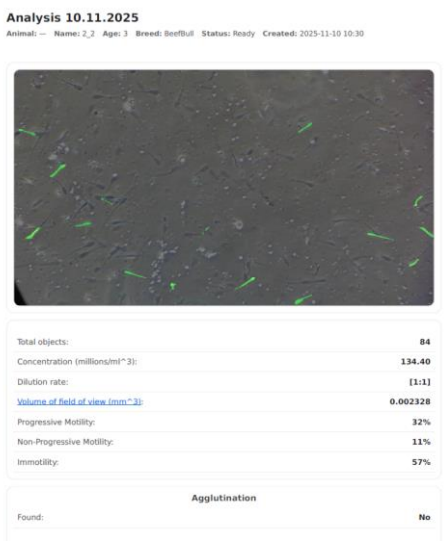
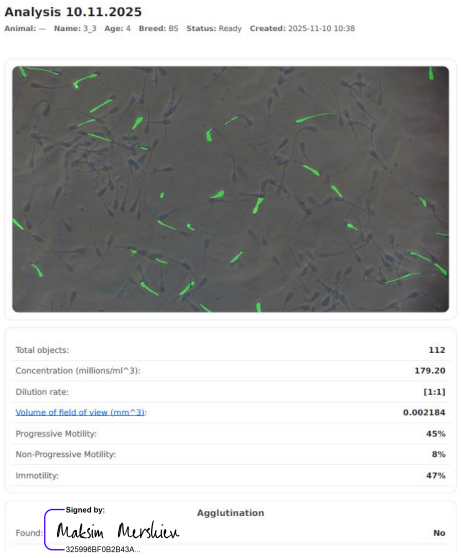
5. Recommendations for Avare

- Standardize dilution and slide preparation during future tests to improve alignment of absolute values.
- Add more species modules (sheep, goats, rabbits, poultry).
- Enable cloud history to compare repeated tests of the same animal.
- Add an **in-app camera for easier recording** and auto-upload.
- Replace error messages with helpful suggestions (adjust focus, dilution, lighting).

Appendix

Screenshot report samples

Bull 1	Bull 2
	
Bull 3	**Bull 3**
	

Analysis 10.11.2025

Animal: — Name: 1_2 Age: 2 Breed: Zavot Status: Ready Created: 2025-11-10 10:19

Total objects:	81
Concentration (millions/ml^3):	129.60
Dilution rate:	[1:1]
Volume of field of view (mm^3):	0.002788
Progressive Motility:	55%
Non-Progressive Motility:	19%
Immotility:	26%

Agglutination

Analysis 10.11.2025

Animal: — Name: 2_2 Age: 3 Breed: BeefBull Status: Ready Created: 2025-11-10 10:30

Total objects:	84
Concentration (millions/ml^3):	134.40
Dilution rate:	[1:1]
Volume of field of view (mm^3):	0.002328
Progressive Motility:	32%
Non-Progressive Motility:	11%
Immotility:	57%

Agglutination

Found:	No

Analysis 10.11.2025

Animal: — Name: 3_3 Age: 4 Breed: BS Status: Ready Created: 2025-11-10 10:38

Total objects:	112
Concentration (millions/ml^3):	179.20
Dilution rate:	[1:1]
Volume of field of view (mm^3):	0.002184
Progressive Motility:	45%
Non-Progressive Motility:	8%
Immotility:	47%

Agglutination

Found: Signed by: Maksim Mershiev 325996BF0B2B43A... No

Analysis 10.11.2025

Animal: — Name: 3_1 Age: 4 Breed: BS Status: Ready Created: 2025-11-10 10:34

Total objects:	179
Concentration (millions/ml^3):	286.40
Dilution rate:	[1:1]
Volume of field of view (mm^3):	0.004869
Progressive Motility:	44%
Non-Progressive Motility:	7%
Immotility:	49%

Agglutination

Found:	No